

TABLE 1–2 Examples of Measurement Scales			
Nominal-level data	Ordinal-level data	Interval-level data	Ratio-level data
Zip code Gender (male, female) Eye color (blue, brown, green, hazel) Political affiliation Religious affiliation Major field (mathematics, computers, etc.) Nationality	Grade (A, B, C, D, F) Judging (first place, second place, etc.) Rating scale (poor, good, excellent) Ranking of tennis players	SAT score IQ Temperature	Height Weight Time Salary Age

EXAMPLE 1–4 Measurement Levels

What level of measurement would be used to measure each variable?

- The ages of patients in a local hospital
- The ratings of movies released this month
- Colors of athletic shirts sold by Oak Park Health Club
- Temperatures of hot tubs in local health clubs

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SOLUTION

- Ratio
- Ordinal
- Nominal
- Interval

Exercises 1–2

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For Exercises 5–7, determine whether the data are qualitative or quantitative.

- Color of football uniforms
- Pizza sizes (small, medium, and large)
- Cholesterol counts for individuals

For Exercises 11–14, determine whether the data are discrete or continuous

- 11.** Number of text messages sent in a month
- 12.** Systolic blood pressure readings
- 13.** Heights of basketball players on a local high school team
- 14.** Number of phone calls received during one day at a poison control center

For Exercises 17–20, give the boundaries of each value.

- 17.** 7 feet
- 18.** 6.3 millimeters
- 19.** 143 miles
- 20.** 3.25 pounds

For Exercises 25–30, classify each as nominal-level, ordinal-level, interval-level, or ratio-level measurement.

- 25.** Amounts of money spent on a medical checkup
- 26.** Scores on a statistical final exam
- 27.** Ratings of fiction books—excellent, good, fair, poor
- 28.** Blood types—O, A, B, AB
- 29.** Online spending
- 30.** Number of flowers in a bouquet

Data Collection and Sampling Techniques

Sampling in Statistics:

Sampling involves choosing a representative group from a population to infer trends, test hypotheses, or make decisions without examining every member. Examples include random sampling, stratified sampling, and cluster sampling.

Random Sampling

A random sample is a sample in which all members of the population have an equal chance of being selected.

Example: Imagine you want to survey the opinions of students about a new cafeteria menu at a university with 2,000 students. Here's how you could perform a simple random sample:

Assign Numbers: Give each student a unique number from 1 to 2,000.

Random Selection: Use a random number generator to select 200 unique numbers.

Form the Sample: The students corresponding to these numbers become your sample for the survey.

Systematic Sampling

A systematic sample is a sample obtained by selecting every k^{th} member of the population where k is a counting number.

Example of Systematic Sampling:

Imagine you have a list of 1,000 employees and you want to select a sample of 100 employees for a survey.

1. **Determine the Sampling Interval:**

Divide the total number of employees by the desired sample size:

$$\text{Interval} = 1000 / 100 = 10$$

2. **Random Starting Point:**

Randomly choose a number between 1 and 10. Suppose you pick 4.

3. **Select the Sample:**

Starting at employee number 4, select every 10th employee. This means you'll pick employee 4, 14, 24, 34, and so on until you have 100 employees.

10 20 30 40

Stratified Sampling

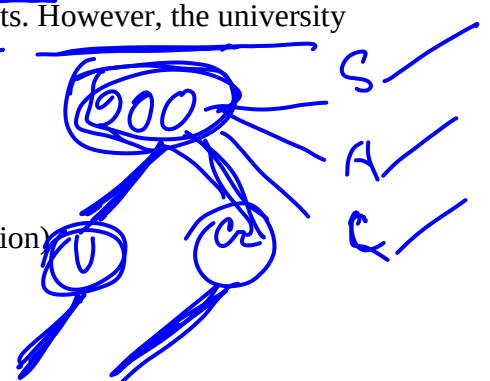
A **stratified sample** is a sample obtained by dividing the population into subgroups or strata according to some characteristic relevant to the study. (There can be several subgroups.) Then subjects are selected from each subgroup.

Example of Stratified Sampling:

Imagine you want to survey 200 university students about their study habits, and you know the student population consists of both undergraduate and graduate students. However, the university population is 80% undergraduate and 20% graduate.

Steps:

1. **Divide the Population into Strata:**
 - Stratum 1: Undergraduate students (80% of the population)
 - Stratum 2: Graduate students (20% of the population)
2. **Determine the Sample Size for Each Stratum:**
 - For undergraduates: $200 \times 0.80 = 160$ students
 - For graduates: $200 \times 0.20 = 40$ students
3. **Randomly Select Within Each Stratum:**
 - Randomly pick 160 undergraduate students from the list of all undergraduates.
 - Randomly pick 40 graduate students from the list of all graduates.



Cluster Sampling

A **cluster sample** is obtained by dividing the population into sections or clusters and then selecting one or more clusters and using all members in the cluster(s) as the members of the sample.

Example of Cluster Sampling:

Imagine you want to survey the academic performance of students in a large city with 50 schools.

1. **Identify Clusters:**
Consider each school as a cluster.
2. **Randomly Select Clusters:**
Randomly choose 10 out of the 50 schools.
3. **Survey Within Selected Clusters:**
Collect data from all students in the selected 10 schools (or you could randomly sample students within these schools if needed).

